

Financial Operating System for Nigerian Informal Businesses

Product concept, MVP, architecture, monetization, regulatory considerations and growth roadmap

A practical blueprint for building a digital back office for Nigerian microbusinesses.

Prepared: September 21, 2026

Executive Summary

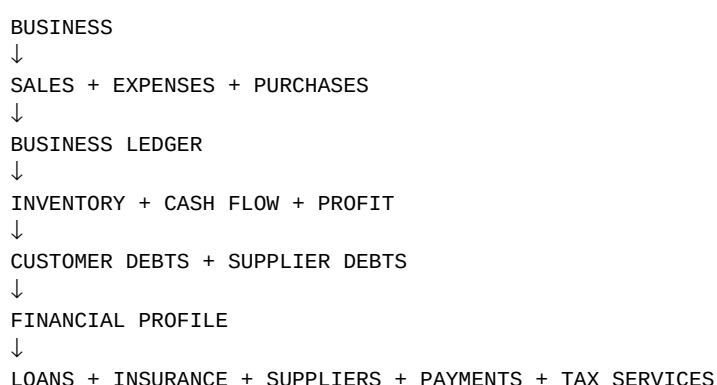
The proposed product is a digital business operating system for Nigerian microbusinesses. Rather than starting as another wallet or payment app, it would help merchants record and understand sales, expenses, inventory, customer debts, supplier debts, profit and cash flow. Over time, the structured business history could support integrations with payments, banking, lending, insurance, suppliers, accounting and tax services.

Core proposition: “Run your entire business from your phone.”

The long-term vision is a financial operating system, but the initial product should solve a concrete operational problem first.

1. What You Are Building

Think of the product as the digital back office of a Nigerian microbusiness.



The business ledger is the foundation. Every important action should create structured financial and operational data.

2. Recommended MVP

Keep the first version small. The MVP should let a merchant operate the essential parts of a business without needing accounting expertise.

Module	Core capability
Dashboard	Revenue, expenses, profit, cash, debts and inventory value
Sales	Record sales, payment method and credit transactions
Inventory	Products, quantities, cost price, selling price and stock value
Expenses	Record operating expenses and categories
Customers	Customer profiles, balances and payment history
Customer debt	Track outstanding balances and repayment history

3. Sales and Automatic Inventory

A sale should not merely become a sales record. It should automatically update inventory and the financial ledger.

Example:

Sale: 2 chargers × ₦9,000 = ₦18,000

Inventory: charger stock 20 → 18

Payment: cash / transfer / POS / credit

If credit: customer receivable increases by the unpaid amount.

4. Inventory Management

Each product can store stock quantity, cost price, selling price, stock value, units sold and estimated gross profit.

Example product:

Oraimo Charger

Stock: 18 units

Cost price: ₦6,000

Selling price: ₦9,000

Stock value: ₦108,000

Units sold: 42

Revenue: ₦378,000

Estimated gross profit: ₦126,000

5. Expenses and Receipt Intelligence

Expense entry should be extremely fast. A later AI feature could allow a merchant to photograph a receipt, extract the merchant, amount, date and category, then ask the user to confirm the transaction.

6. Customer Credit / Debt Management

Customer debt can be a major feature because informal merchants often sell on credit. The system should track individual transactions, outstanding balance, repayments and reminders.

Customer: Chinedu Okafor

Outstanding: ₦72,000

₦25,000 – 04 Sept

₦17,000 – 09 Sept

₦30,000 – 14 Sept

Next reminder: 18 Sept

7. Supplier Management

The same principle should apply to suppliers: purchases, amounts owed, purchase history and upcoming obligations.

8. The Business Financial Graph

After months of use, the platform can build a structured representation of how a business operates: revenue, margins, inventory, customers, suppliers, expenses, purchasing patterns and payment history.

Example business profile:

Monthly revenue: ₦3.2m

Monthly expenses: ₦2.4m

Gross margin: 25%

Inventory: ₦4.8m

Customers: 312

Average daily sales: ₦105k

Customer credit: ₦340k

Supplier purchases: ₦1.8m/month

Business history: 18 months

9. Embedded Financial Services

Once the platform has reliable business records, it can potentially connect merchants to licensed partners for financing, insurance, payments, banking and supplier services. The software company does not necessarily need to become the lender, insurer or bank.

```
BUSINESS DATA
↓
FINANCIAL PROFILE
↓
LICENSED PARTNERS
■■■ Financing
■■■ Insurance
■■■ Payment services
■■■ Banking / account services
```

10. Tax and Formalisation

Tax functionality should initially be positioned as a consequence of good recordkeeping rather than the main selling point. The product can organize transaction records and eventually integrate with appropriate tax/accounting services.

11. Technology Architecture

A practical initial stack could be:

```
Mobile: React Native + Expo + TypeScript
Navigation: Expo Router
State: Zustand
API client: Axios
Backend: Node.js / NestJS
Database: PostgreSQL
Cache / queues: Redis
Cloud: AWS, Azure or GCP
Realtime: WebSocket where useful
```

The exact stack can change. The critical architectural decision is to make the ledger and synchronization model reliable.

12. Database Philosophy

Design around financial and operational events rather than around screens. This makes reporting, reconciliation, auditing and future integrations easier.

```
SALE_CREATED → ■25,000
PAYMENT_RECEIVED → ■15,000
CUSTOMER_CREDIT_CREATED → ■10,000
INVENTORY_DECREASED → 2 units
```

13. Offline-First Design

Offline capability is important. A merchant should be able to record a sale without an active internet connection. Transactions can be stored locally and synchronized with the cloud when connectivity returns.

```
Mobile local database
↓
Sync engine
↓
Cloud API
```

↓
PostgreSQL

14. WhatsApp as an Interface

WhatsApp can reduce friction for customer debt reminders and, later, potentially support conversational transaction entry.

15. Practical AI Features

AI should be applied after reliable structured transaction data exists. Useful applications include receipt extraction, daily summaries, inventory forecasts, unusual-expense detection, cash-flow warnings and natural-language business questions.

Merchant: "How much did I make selling chargers last month?"

System: "You generated ₦842,000 in charger sales, with an estimated gross profit of ₦214,000."

16. Initial Target Market

Do not launch for every type of informal business simultaneously. A focused first vertical could be phone/accessories retailers because they have many SKUs, frequent transactions, inventory needs, supplier relationships and customer-credit use cases.

Possible expansion: phone accessories → fashion → beauty/cosmetics → electronics → spare parts → wholesalers → restaurants → small manufacturers.

17. Monetisation

A freemium SaaS model can provide basic sales, inventory and expense tracking, with paid tiers for advanced reports, employees, multiple branches, backups, analytics and AI. Later revenue could also come from financial-service and supplier partnerships, subject to the applicable commercial and regulatory arrangements.

18. Regulatory Architecture

There is an important distinction between recording business finances and performing regulated financial activities such as holding funds, providing payments or lending. The initial product can focus on software and data while integrating regulated services through appropriately licensed partners.

Because the platform may process personal and financial information, privacy, security, access controls, audit logging, consent, data minimization, retention and incident-response processes should be considered from the beginning. Nigerian data-protection requirements should be reviewed with qualified legal/compliance professionals before launch.

19. Development Roadmap

Stage	Focus	Key deliverables
1	Merchant MVP	Registration, business profile, products, inventory, sales, expenses, customers, debt and dashboard
2	Business management	Suppliers, purchases, supplier debt, profit, cash flow, receipts, exports and offline mode
3	Scale	Employees, roles, multiple devices/branches, barcode scanning, WhatsApp and analytics

Stage	Focus	Key deliverables
4	Financial infrastructure	Bank feeds, payment integrations, reconciliation and partner integrations
5	Financial identity	Business history, revenue, inventory, cash flow and other structured indicators
6	Ecosystem	Financing, insurance, supplier marketplace and broader business services through partners

20. Potential Product Experience

HOME

Revenue this month: ₦4,820,000

Gross profit: ₦1,120,000

Expenses: ₦740,000

Cash: ₦680,000

Inventory: ₦3,840,000

THINGS YOU SHOULD KNOW

- 3 products may run out within 7 days
- Customer debts increased 22% this month
- Revenue increased 14%
- Supplier payment of ₦420k due Friday

21. The Long-Term Moat

The interface itself is not the moat. Potential defensibility comes from accumulated transaction history, merchant relationships, structured business intelligence, integrations with financial infrastructure and switching costs created by years of business records.

```

Business records
↓
Business management
↓
Financial intelligence
↓
Financial identity
↓
Embedded financial services
↓
Business ecosystem

```

22. Recommended First Experiment

Before investing in a large engineering build, interview approximately 30–50 real Nigerian informal merchants. Test the problem, not the technology. Ask what they currently struggle to know about their business and observe how they record sales, expenses, inventory and debts.

A particularly important validation question is whether merchants will consistently record enough accurate transactions for the product to become useful. The CBN-published research referenced in the original concept discusses barriers including digital literacy, trust and incentives, reinforcing the importance of tangible value to merchants.

23. Suggested North-Star Product

Short-term promise: Run your entire business from your phone.

Long-term vision: Become the financial operating system and business infrastructure layer for Nigerian

microbusinesses.

Sources and Further Reading

CBN-published research: “Digital Financial Services and Informal Sector Formalisation in Nigeria” — Central Bank of Nigeria Bullion, Vol. 50, Issue 2 (2026).

Nigeria Data Protection Commission: Nigeria Data Protection Act 2023 and related regulatory information.

Note: Regulatory, tax, lending, payments and financial-services decisions should be validated against current Nigerian requirements and with qualified professional advisers before implementation.

Addendum, October 2026

Product status, feature roadmap, version plan and pricing by role

Prepared: October 7, 2026. Appended to the original blueprint of September 21, 2026; sections 1–23 above are unchanged.

Since the blueprint was written, BizLedger has been built and deployed: a NestJS API on Render backed by PostgreSQL on Supabase, and an Android app built with React Native and Expo. This addendum records what the product does today (release v1.1.0), consolidates every feature still planned, from this blueprint, the **BizLedger Feature Expansion Documentation & Product Roadmap** (October 2026), the **Smart SME Financing and Growth Platform** pitch (October 2026) and the production-readiness review, into one versioned roadmap, and proposes what can reasonably be charged for each role.

24. What BizLedger Does Today (v1.1.0)

Everything below is built, deployed and working in the current release.

Module	Current capability
Accounts and onboarding	Three-slide onboarding; business registration with business name, owner name and phone; an 18+ confirmation recorded at sign-up; email and password login; the session stays signed in on the device; logout behind a confirmation.
Roles and team	Owner and staff accounts. Owners create and remove staff and assign them to a branch. Staff can sell, add products and manage customers, but never see costs, profit, expenses, analytics or history; this is enforced by the server, not just hidden in the app. Removing a staff member takes effect within about 30 seconds.
Branches	Multiple branches with addresses. Sales and expenses are tagged to the branch they happened in; owners view the dashboard and analytics for all branches or one. Stock is one shared pool per business.
Products and inventory	Products with selling price, cost price, SKU, barcode, low-stock alert level and an optional photo (camera or gallery, resized on the phone). Search and Low/Out-of-stock filters, stock value at cost, stock adjustments that require a reason (restock, damaged or lost, count correction, customer return), and a stock history per product. Barcode scanning when adding stock and at the point of sale. A product with sales cannot be deleted.
Sales	Multi-item cart with quantity steppers, scan-to-add and a per-line price override for haggling (the cost basis is kept for true profit). Cash, transfer, POS or credit; transfers record the channel (bank, OPay, PalmPay, other) and a reference; credit sales record the amount paid now. Stock is deducted automatically. Sales list with today and this-month totals, search and Paid/Owing/Unverified filters.
Customers and credit	Customer profiles with phone, purchase summary and outstanding balance. Repayments are checked against the balance and allocated oldest-first across that customer's credit sales. Call or WhatsApp from the profile, WhatsApp debt reminders, an activity timeline, edit, and a delete that is refused while the customer has history.
Expenses	Seven categories (rent, transport, salary, utilities, supplies, maintenance, other) with notes and branch tagging; this-month and all-time totals; a category breakdown; search and category filter; edit with a change history; deletes kept on record.
Dashboard	Greeting by name; revenue, gross and net profit, expenses, cash in hand, inventory value and customer count; money owed by customers; a 7-day revenue and expense chart; top products; plain-language insights; a running-low list; recent activity.
Analytics	7 days, 30 days, 3 months or 12 months, by day, week or month: sales trend, expenses by category, best sellers, slow movers, stock movement, customer insights (active, new, returning, best customers) and payment-method mix; filterable by branch.

Module	Current capability
Event ledger and activity	Every change (sale, payment, stock movement, product, customer and expense create/edit/delete) is written as an append-only event in the same database transaction, with the item and the person who did it. An activity feed shows the whole business by day, and each product, customer and expense has its own history.
WhatsApp	A prefilled receipt after every sale and debt reminders from the customer screen, through WhatsApp links; no WhatsApp Business account needed.
Platform and security	Versioned database migrations; Row Level Security on every table; every query scoped to the signed-in business; server-side role checks and cost redaction for staff; a required production secret; closed CORS; compressed responses; timestamps handled in UTC; the app shows a banner while the hosted server wakes up.
Releases	One version number for the API and the app (now 1.1.0), a changelog, git tags for every release, and /health reporting the running version.

Progress against the original roadmap (section 19)

Stage	Status	Notes
1 Merchant MVP	Complete	All six MVP modules from section 2 are built.
2 Business management	Partly complete	Profit, cash flow and analytics are built. Suppliers, purchases, supplier debt, receipts, exports and offline mode are not.
3 Scale	Mostly complete, ahead of order	Employees, roles, branches, barcode scanning, WhatsApp and analytics are built. Multiple devices work through the shared online account; per-branch stock is not built.
4–6	Not started	Bank feeds, payment integrations, financial identity and the partner ecosystem.

25. Known Gaps in the Current Release

- No suppliers, purchases or supplier debt; no draft, cancelled or refunded sales (the database is ready for them, the app is not).
- No offline mode: recording a sale needs a connection.
- No receipts or statements as PDF, and no CSV or Excel export or import.
- No password reset, no rate limiting on sign-in, a 6-character minimum password, and the session token is kept in ordinary app storage rather than the phone's secure keychain.
- No way to retire old app versions yet (a minimum-version check is designed for v1.2).
- No privacy policy, terms of service or NDPA review; no backup plan beyond the hosting provider's defaults.
- Two colours fail accessibility contrast (the warning orange and the placeholder grey).
- No AI features, scoring, financial profile, financing partners or payment integrations yet.

26. Features To Be Added

Consolidated from five sources: **B** = this blueprint (Sept 2026); **F** = Feature Expansion Documentation & Product Roadmap (Oct 2026); **P** = Smart SME Financing pitch (Oct 2026); **R** = production-readiness review and release planning (Sept–Oct 2026); **O** = operations and infrastructure plan (Oct 2026).

Area	Feature	Source
Trust and foundation	Minimum supported app version with an in-app "update required" screen; password reset and a stronger password policy; sign-in rate limiting; secure token storage; security headers; accessible colours; privacy policy, terms and NDPA review; backup and recovery plan.	R
Supply side	Suppliers, purchase orders and goods receipts, so each supplier's real lead time can be measured; supplier debt with payment history and upcoming obligations; cancel and refund sales; per-branch stock (optional).	B, F, O
Digital twin	Per-product stock, demand, supplier and stockout-risk figures rebuilt from the ledger. Alerts and recommendations rank alternative local suppliers and let the merchant draft and approve a purchase order in a couple of taps. The first forecasts are statistical, not machine-learned.	O
Composable micro-ERP	A module registry with on/off flags for each business, plus operating-model presets (wholesale, retail, direct-to-consumer). Built as a modular monolith, not microservices.	O
Scenario planning	"What if" questions answered by a deterministic simulator that computes every number. The language model only turns the question into parameters and writes the contingency steps around the computed results, and a checker rejects any number in the text that the simulator did not produce.	O
Harden and ship	Offline capture; review of data handling against the Nigeria Data Protection Act by counsel; production monitoring; a pilot with real merchants.	B, R, O
Records and documents	PDF receipts and statements; CSV and Excel export; automated daily summaries, weekly performance reports, monthly management accounts, quarterly business-health reports and lender/investor-ready business packages.	B, F
Data capture	Offline-first recording with sync; CSV and Excel import; bank-statement upload; receipt photo capture with AI extraction; later bank, POS and payment-gateway feeds.	B, F
Business intelligence	A metrics engine (revenue, margins, cash flow, receivables, payables, working capital) that explains changes in words, such as "profit margin fell from 24% to 19%"; cash-flow warnings, unusual-expense detection and inventory forecasts.	B, F
AI copilot and advisers	Questions answered from the business's own data ("Why did my profit fall?", "Can I afford a loan?"); a growth adviser with prioritised actions; an AI funding adviser that checks affordability before showing any financing.	B, F, P
Business health and credit readiness	BizHealth: a continuously updated health score out of 100 with category scores and corrective actions. BizScore: an explainable, rules-based credit-readiness score with a data-confidence indicator, later moving to machine-learning models. A financing-readiness percentage that says what is holding it back. A business maturity ladder from "Business Activity" to "Scale Ready".	F, P
Business identity	BizPassport: a portable business profile with history, BizHealth, BizScore and verification status; a document vault for CAC, TIN, bank statements, tax documents, invoices, permits, insurance and contracts.	B, F
Permissioned data sharing	Banks, fintechs and DFIs request specific data for a stated purpose; the SME approves the scope and duration; every access is logged, expires and can be revoked.	F

Area	Feature	Source
Capital marketplace	Matching to working-capital, inventory, asset, invoice, supplier and revenue-based financing and eligible grant or DFI programmes, including programmes for women-led businesses; amounts, fees, tenor, collateral and eligibility shown up front; a partner dashboard. BizLedger matches and refers; licensed partners lend.	B, F, P
Growth marketplace	Supplier discovery and price comparison, wholesale and distribution opportunities, tenders, training and business-development resources; insurance through licensed partners.	B, F, P
Financing-need diagnosis	Before any matching, work out what the money is for (working capital, inventory, equipment or other assets, invoice or supplier financing) from the stated purpose and the business's own records, and recommend the type, amount range and tenor its cash flow can support.	P
Fair assessment	BizHealth, BizScore and financing matches use business and financial behaviour only, never gender, ethnicity, religion, age or other demographic proxies. Every factor is documented and explainable, and outcomes are checked regularly for bias.	P
Financial education and guidance	Short in-app lessons and cash-flow guidance built on the business's own numbers (pricing and margins, managing customer credit, saving for restock, preparing for a loan), with programmes and resources for women-led businesses shown when the business is eligible.	P
Post-financing tracking	Record each loan received and its repayment schedule; track repayments with due-date reminders; measure what changed after funding (inventory, sales, cash flow); and feed repayment behaviour back into BizHealth, BizScore and the financial profile.	P
Analytics and AI stack	A Python analytics service (Pandas, scikit-learn) beside the NestJS API for metrics and scoring; the OpenAI API for the copilot and plain-language explanations; gradient-boosted models such as XGBoost once enough history exists.	P
Channels	WhatsApp transaction entry ("Sold 4 chargers for 60k cash") that returns a confirmation and the stock and financial impact; a web app for owners, accountants and partners.	B, F
Integrations and scale	Bank feeds, POS, Paystack and Flutterwave, reconciliation of payments to sales, accounting and tax services, e-commerce and supply-chain integrations, and a machine-learning credit engine trained on alternative data.	B, F, P

27. Version Roadmap

Each release keeps the single version number adopted in v1.1.0: minor releases add features, major releases mark a new product generation (Intelligence, Financial Identity, Financing Infrastructure, as in the Feature Expansion document). Following its strategic principle, new capabilities are built on one central business-data and metrics layer beneath the app, so the same transactions power the dashboard, the AI, BizHealth, BizScore, BizPassport and the financing engine. Versions 1.3 to 1.7 follow the five-step operations plan in section 28, in order: the supply side comes first because the digital twin cannot measure supplier lead times without purchase orders and goods receipts. Target dates are estimates for a small team and assume the merchant interviews in section 22 confirm demand.

Version	Theme	Key deliverables	Target
1.1.0	Revamp (released)	Design system, editable records with history, product photos, analytics by period, activity feed, event ledger for every change.	Oct 2026
1.2	Trust and updates	Minimum app version and update screen; password reset; rate limiting; stronger passwords; secure token storage; security headers; contrast fixes; privacy policy and terms; backups.	Nov 2026
1.3	Supply side	Suppliers, purchase orders and goods receipts, with each supplier's measured lead time; supplier debt and payments; cancel and refund sales; PDF receipts and statements; CSV and Excel export.	Dec 2026
1.4	Digital twin	Per-product stock, demand, supplier and stockout-risk figures rebuilt from the ledger; stockout alerts; ranked alternative local suppliers; draft and approve a purchase order in a couple of taps; statistical forecasts.	Feb 2027
1.5	Composable micro-ERP	Module registry with per-business on/off flags; wholesale, retail and direct-to-consumer presets; storefront orders; still one modular monolith.	Mar 2027
1.6	Scenario planning	Deterministic "what if" simulator; plain-language questions turned into parameters; contingency steps written around computed results; a checker that rejects any number the simulator did not produce.	Apr 2027
1.7	Harden and ship	Offline capture with sync; Nigeria Data Protection Act review by counsel; monitoring, alerting and backups; the early-users infrastructure (section 29); a pilot with real merchants.	Jun 2027
2.0	BizLedger Intelligence	Metrics engine with plain-language explanations; automated daily, weekly and monthly reports; AI business copilot; growth adviser; cash-flow warnings and unusual-expense alerts; financial education lessons and cash-flow guidance; CSV and Excel import, bank-statement upload and receipt photo capture; a Python analytics service and OpenAI-powered explanations.	Aug 2027
2.1	BizHealth	Business health score with category scores, risk indicators, recommendations and the maturity ladder.	Sep 2027
2.2	BizScore and readiness	Explainable credit-readiness score with confidence indicator; financing-readiness percentage; AI funding adviser ("Can I afford this loan?"); financing-need diagnosis; fairness rules and bias checks for every score.	Nov 2027
3.0	Financial Identity	BizPassport; document vault and verification; permissioned data sharing with consent, scopes, expiry, audit log and revocation; web app and partner portal.	Jan 2028
3.1	WhatsApp entry	Conversational transaction recording through WhatsApp with confirmations.	Feb 2028
4.0	Financing Infrastructure	Financing API and marketplace with partner listings, eligibility engine, matching, loan calculator, applications and partner dashboard; supplier financing; programmes for women-led businesses; loan and repayment tracking after funding, with outcomes fed back into the profile.	Apr 2028
4.x	Integrations and scale	Bank feeds, POS and payment-gateway integrations, reconciliation, insurance, supplier and opportunity marketplace, tax and accounting services, machine-learning credit engine; the growth-stage infrastructure (section 29).	2028–2029

Hackathon slice. For the Smart SME Financing track, a thin demonstration of the full data-to-capital journey (transactions → business intelligence → AI insight → BizHealth → BizScore → BizPassport → financing match with mock partners) can be built ahead of this schedule, as the Feature Expansion document recommends, without changing the production order above.

28. Operations Intelligence Plan

Five steps, built in this order between v1.3 and v1.7. Each step names the test that shows it is done.

Step 1. Supply side (v1.3)

Suppliers, purchase orders and goods receipts. A purchase order records what was ordered, from which supplier, at what price and when it is expected; a goods receipt records what actually arrived and when, and adds it to stock through the ledger. The gap between the two is the supplier's real lead time, which the digital twin cannot measure without them. Supplier debt and payments are recorded alongside, completing the payables side of the ledger.

Done when: a merchant can raise a purchase order, receive goods against it in full or in part, and see each supplier's measured lead time.

Step 2. Digital twin (v1.4)

A live model of every product, rebuilt from the ledger: stock on hand, demand rate, days of cover, supplier lead time and reliability, and stockout risk. When a product is at risk, BizLedger alerts the merchant, recommends a reorder quantity, ranks alternative local suppliers by price, lead time and reliability, and drafts a purchase order the merchant approves in a couple of taps. The first forecasts are statistical, built from the merchant's own sales history (for example moving averages and weekly patterns), not machine-learned; models come later, once enough history exists.

Done when: every product shows days of cover and stockout risk, and a risk alert leads to an approved purchase order in a couple of taps.

Step 3. Composable micro-ERP (v1.5)

A module registry with on/off flags for each business (for example inventory, purchasing, customer credit, branches and storefront), and operating-model presets that switch on the right modules for wholesale, retail or direct-to-consumer (D2C) businesses. BizLedger stays a modular monolith: one deployable API with clear module boundaries, not microservices.

Done when: a wholesale merchant can be switched to D2C and take a storefront order with no data migration and no redeploy.

Step 4. Scenario planning (v1.6)

Merchants ask "what if" questions in plain language, such as "What if my main supplier raises prices by 15%?" or "What if sales fall by a fifth in January?". A deterministic simulator, working from the digital twin and the ledger, computes every number. The language model only turns the question into simulator parameters and writes the contingency steps around the computed results. A checker compares every number in the written answer with the simulator's output and rejects the answer if any number did not come from the simulator, so the AI cannot invent figures.

Done when: every figure in a scenario answer can be traced to a simulator output, and answers containing any other number are rejected automatically.

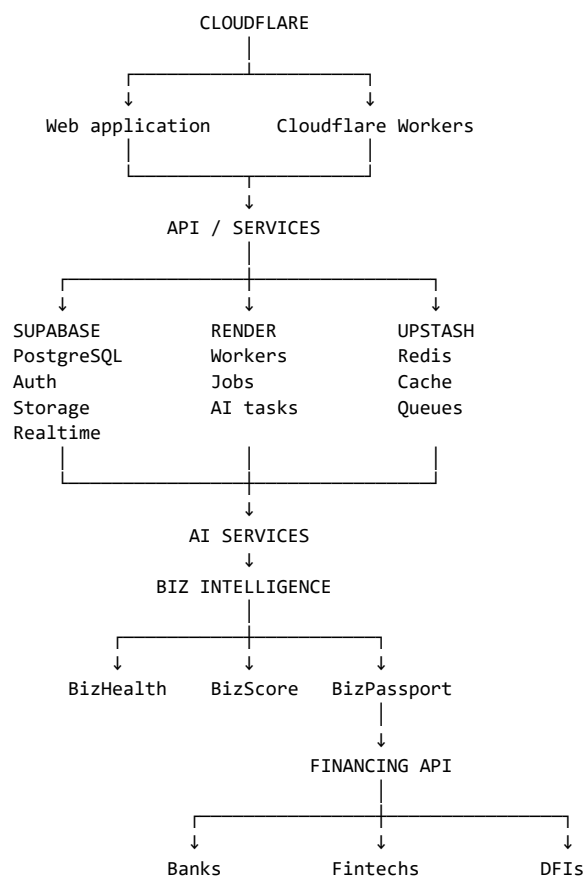
Step 5. Harden and ship (v1.7)

Offline capture, so sales and stock can be recorded without a connection and synchronised when it returns; a review of data handling against the Nigeria Data Protection Act by qualified counsel; production monitoring, alerting and backups; and a pilot with real merchants to test whether they record consistently and act on the alerts.

Done when: pilot merchants run their businesses on BizLedger through a defined pilot period, counsel's recommendations are in place, and incidents are caught by monitoring before merchants report them.

29. Target Infrastructure Architecture

The platform as it grows into the full roadmap. Cloudflare sits in front of the web application and runs edge Workers; both reach the API and services. These use Supabase (PostgreSQL, authentication, file storage and realtime updates), Render (the API, background workers, scheduled jobs and AI tasks) and Upstash Redis (caching and queues). AI services feed the business-intelligence layer that produces BizHealth, BizScore and BizPassport, and BizPassport reaches banks, fintechs and DFIs through the financing API, only with the business's consent.



Today vs target. Part of this exists now: the API runs on Render and the database on Supabase PostgreSQL. Authentication is handled inside the API and product photos are stored in the database; Cloudflare, Upstash, Supabase Auth and Storage, the AI services and the financing API are the target. Whatever the hosting, the API stays the modular monolith described in section 28.

Infrastructure by stage

Stage	When	Infrastructure	Also introduce
Today	Building and testing, before real merchants	Render web service (free tier, which sleeps when idle) and Supabase PostgreSQL (free tier, 15 pooled connections); Android app built with EAS	Nothing more until the pilot
Early users	Roughly hundreds to thousands of active businesses	Supabase Pro, Render, Cloudflare, Upstash	Proper monitoring and backups
Growth stage	Significant transaction volume	Supabase, Cloudflare, Render, Upstash	Dedicated AI and data-processing infrastructure; bank and payment integrations

30. Roles and What Each Can Do

Today the product has two roles, owner and staff. The roadmap adds three: a branch manager, an accountant or adviser, and external partners who see data only with the business's consent.

Capability	Owner	Staff	Branch manager (planned)	Accountant (planned)	Lender partner (planned)
Record sales and take repayments	Yes	Yes	Yes	No	No
Add products and customers	Yes	Yes	Yes	No	No
Edit products, prices and stock	Yes	No	Own branch	No	No
See costs, profit and expenses	Yes	No	Own branch	Read only	With consent
Record and edit expenses	Yes	No	Own branch	No	No
Dashboard, analytics, history	Yes	No	Own branch	Read only	With consent
Manage staff and branches	Yes	No	No	No	No
Draft and approve purchase orders (v1.3+)	Yes	No	Own branch	Read only	No
Reports and exports (v1.3+)	Yes	No	Own branch	Yes	With consent
Stockout alerts and supplier ranking (v1.4+)	Yes	Alerts only	Own branch	Read only	No
Turn modules on or off (v1.5+)	Yes	No	No	No	No
Scenario planning (v1.6+)	Yes	No	Own branch	Yes	No
AI copilot and BizHealth (v2+)	Yes	No	Own branch	Read only	No
BizScore and BizPassport (v2.2+)	Yes	No	No	Read only	With consent
Approve data-sharing requests (v3+)	Yes	No	No	No	Requests only
Make financing offers (v4+)	No	No	No	No	Yes

31. Pricing by Role

These are proposed prices, not market research, and should be tested in the merchant interviews recommended in section 22. They follow the freemium model in section 17: recording stays free so merchants build history, and paid tiers unlock team, branch, intelligence and financing features. Every price is in naira, per month, with annual billing at ten months' price.

Merchant plans (paid by the business owner)

Plan	Price	Includes
Starter	Free	One owner; up to 50 products; sales, inventory, expenses, customers and debt; dashboard; WhatsApp receipts and reminders. Enough to prove value and start building history.
Growth	₦3,500 / month	Unlimited products and photos; full analytics and activity history; exports and PDF receipts (from v1.3); one staff seat included.
Business	₦9,500 / month	Everything in Growth, plus up to 3 branches, 5 staff seats, branch reports, suppliers and purchase orders (from v1.3), digital-twin stockout alerts and supplier ranking (from v1.4), automated weekly and monthly reports and priority support.
Intelligence (from v1.6)	₦15,000 / month	Everything in Business, plus scenario planning (from v1.6), the AI copilot, BizHealth, BizScore, financing readiness and, from v3.0, BizPassport and the document vault.

Additional seats, by role

Role	Price	Rationale
Staff	₦1,000 / month each	Sales-only access; low enough that owners give every cashier their own login instead of sharing the owner's.
Branch manager (planned)	₦1,500 / month each	Adds stock, expense and reporting control for one branch.
Extra branch	₦3,000 / month each	Beyond the 3 branches in the Business plan; covers per-branch reporting.
Accountant or adviser (planned)	Free for one business; ₦7,500 / month for up to 10 client businesses	Accountants bring many merchants with them, so access is priced to encourage them to onboard clients.

Module add-ons (from v1.5)

Module	Price	Notes
Storefront (direct-to-consumer)	₦2,500 / month	Online storefront orders flow into the same ledger, stock and customer records; switched on per business with no migration.
Wholesale preset	Included in Business	Bulk pricing, purchase orders and customer credit switched on together.

Partner side (paid by financial institutions and suppliers)

Partner role	Proposed charge	Notes
Lender data access (v3.0+)	₦2,000–₦5,000 per consented BizPassport report, or a partner subscription from ₦150,000 / month	Only with the business's approval, for a stated purpose and period, with an audit log.
Financing referral (v4.0+)	1–2% of the amount disbursed, paid by the lender	BizLedger matches and refers; licensed partners lend. Free for the SME, which keeps the matching neutral.
Supplier marketplace (v4.x)	1–3% commission on completed orders	Suppliers pay for orders that come through BizLedger.
DFI and grant programmes	Per-programme platform fee, agreed case by case	For administering eligibility and reach, for example programmes for women-led businesses.

Affordability check. For the pitch's example retailer with ~~₦2.4~~ million in monthly sales, the Growth plan is about 0.15% of monthly sales and the Business plan about 0.4%. The price should be justified by debt recovered and stock losses avoided, not just by convenience. Billing itself is not built yet; it would arrive with v1.2 or v1.3 through a local payment provider. Referral fees, data-access charges and any lending-related revenue must be reviewed against current CBN, FCCPC and NDPA requirements with qualified advisers before launch.

32. Smart SME Financing Challenge Coverage

How the roadmap answers the Smart SME Financing and Growth Platform challenge: using SME business data to improve access to financing and relevant financial products, open new market opportunities and support sustainable growth, including for women-led businesses.

Challenge requirement	How BizLedger meets it	Version
Use alternative business data	Every sale, payment, stock movement and expense is already recorded as an event; the metrics engine turns this into revenue, expense, inventory, receivable, payable and cash-flow behaviour.	Built; 2.0
Show creditworthiness without a black box	BizScore scores revenue consistency, cash-flow stability, inventory performance, debt management and data quality, each with reasons and a confidence indicator.	2.2
Make SMEs finance-ready	A financing-readiness percentage with strengths, gaps and next actions; financing-need diagnosis; the AI funding adviser checks affordability first.	2.2
Access appropriate financing	Marketplace matching to banks, fintechs, microfinance banks, DFIs and grants, with terms shown up front; the business approves what data each lender sees.	3.0; 4.0
Women-led businesses	Eligible programmes surfaced in matching; financial education and cash-flow guidance; fair assessment that never uses demographic proxies.	2.0–4.0
New market opportunities	Growth adviser from product momentum; supplier discovery and price comparison; wholesale, distribution, tender, grant and training opportunities.	2.0; 4.x
Sustainable growth after funding	Loan and repayment tracking, with sales, inventory and cash flow measured after funding and repayment behaviour strengthening the profile.	4.0

Challenge requirement	How BizLedger meets it	Version
Operational resilience	The digital twin flags stockout risk early and ranks alternative local suppliers; scenario planning tests "what if" questions with computed, verifiable numbers.	1.4; 1.6
Hackathon demonstration	One business, one financing need, one complete journey, with mock financing partners.	Ahead of schedule

33. Next Steps

- Ship v1.2 (Trust and updates), including the minimum-version check so older apps can be retired.
- Run the 30–50 merchant interviews from section 22, adding the pricing in section 31 as a willingness-to-pay test.
- Start the supply side (v1.3): suppliers, purchase orders and goods receipts, so lead-time data accumulates before the digital twin is built.
- Write the privacy policy and terms now, and book the Nigeria Data Protection Act review with counsel ahead of the v1.7 pilot.
- Recruit pilot merchants for v1.7 from the interviews, and set up monitoring and backups before they start.
- Define the BizHealth and BizScore factors and weights on paper, so the v2 data layer collects what they need from the start.
- Identify two or three potential financing partners to shape the permissioned data-sharing and marketplace design.

Sources for this addendum

BizLedger source code and release v1.1.0 (git tag, October 2026); CHANGELOG, README and PRODUCTION_TODO in the BizLedger repository; BizLedger Feature Expansion Documentation & Product Roadmap (October 2026); BizLedger "Smart SME Financing and Growth Platform" pitch deck (October 2026); BizLedger operations and infrastructure plan (October 2026); sections 1–23 of this document.

Note: prices, dates and partner arrangements in this addendum are proposals for planning, not commitments, and should be validated with merchants, partners and qualified regulatory advisers.